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Identifying High Risk Groups and Quantifying Absolute Risk of Cancer After Kidney Transplantation: A Cohort Study of 15 183 Recipients

A. C. Webster[a,b,c,d,*], J. C. Craig[a,b], J. M. Simpson[b], M. P. Jones[a] and J. R. Chapman[a,c]

[a] *NHMRC Centre for Clinical Research Excellence in Renal Medicine, Children's Hospital at Westmead, NSW, Australia*
[b] *School of Public Health, University of Sydney, NSW, Australia*
[c] *Centre for Transplant and Renal Research, Westmead Hospital, NSW, Australia*
[d] *Australia and New Zealand Dialysis and Transplant (ANZDATA) Registry, Queen Elizabeth Hospital, Adelaide, Australia*
[*] *Corresponding author: Angela Webster, angela.webster@gmail.com*

Transplant recipients have increased cancer risk, but data on risk variation across different patient groups are sparse. Rates and standardized rate ratios (SRR) of cancer (all sites, excluding nonmelanocytic skin and lip cancer) compared to the general population were calculated, using Australia and New Zealand Dialysis and Transplant Registry data. Within the transplant population, risk factors were identified (hazard ratios: HR; 95% CI) and absolute risk estimated for recipient groups. A total of 1642 (10.8%) of 15 183 recipients developed cancer. Risk was inversely related to age (SRR 15–30 children, 2 if >65 years). Females aged 25–29 had rates equivalent to women aged 55–59 from the general population. Age trend for lymphoma, colorectal and breast risk was similar; melanoma showed less variability across ages, prostate showed no risk increase. Within the transplanted population, risk was affected by age differently for each sex ($p = 0.007$), elevated by prior malignancy (HR 1.40; 1.03–1.89), white race (HR 1.36; 1.12–1.89), but reduced by diabetic end-stage kidney disease (ESKD) (HR 0.67; 0.50–0.89). Cancer rates in kidney recipients are similar to nontransplanted people 20–30 years older, but absolute risk differs across patient groups. Men aged 45–54 surviving 10 years have cancer risks varying from 1 in 13 (non-white, no prior cancer, diabetic ESKD) to 1 in 5 (white, prior cancer, other ESKD).

Key words: Cohort study, kidney transplantation, malignancy, risk

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Background

Kidney transplantation is the treatment of choice for most patients with end-stage kidney disease (ESKD), as transplantation confers survival and quality of life advantages over dialysis (1). Globally, the mortality rate on dialysis is 10–15% per year, compared with 2–4% posttransplantation (2–4). However, kidney transplantation is not without risk, as recipients are at increased risk of cardiovascular disease, cancer and infection and the relative mortality rate for kidney recipients after the first post-transplant year remains 4–6 times that of the general population, with 33% deaths attributable to cardiovascular disease and 32% to cancer (3,5,6).

Elevated cancer risk after transplantation is thought to result from the interplay of several factors: the chronic uraemic state favors carcinogenesis and overall or cumulative exposure to immunosuppression disrupts both antitumor immunosurveillance and anti-viral activity, and may potentiate the carcinogenic effects of other agents such as sunlight (7,8). Additionally, some drugs promote carcinogenesis by mechanisms independent of their immunosuppressive effects (9). Viral infections (including herpes and hepatitis viruses) are clearly linked to some malignancies and chronic antigen stimulation from the transplanted organ, repeated infections and transfusions of blood products have also been implicated.

Quantifying an individual transplant, candidate's cancer risk informs pre-transplant counseling, subsequent treatment decisions and has implications for monitoring, screening and follow-up after transplantation (10). Most previous published analyses of cancer risk after transplantation have concentrated on specific cancers (11–13), or on risk associated with modifiable factors after the fact of transplantation, such as particular components of the immunosuppressive regimen employed (14,15), but have not fully elaborated the underlying epidemiology of malignancy in this population. Many analyses are constrained by their data sources: either insufficient sample size to allow adjustment for all confounders, incomplete population coverage and/or reliance on secondary data sources, all of which may introduce unmeasurable error and potential bias (16–18).

The aims of this study were firstly to establish the relative risk of cancer (excluding nonmelanocytic skin and lip cancer) in the post-transplant population compared to that experienced by the general population, and secondly to quantify how excess risk varied within the transplanted population, seeking to establish meaningful absolute risk estimates for post-transplant cancer based on unalterable recipient characteristics known *a priori* at the time of transplantation.

Methods

Study population

We performed an inception cohort study of all adults and children in the Australia and New Zealand Dialysis and Transplant Registry (ANZDATA) from the time of first kidney transplantation. ANZDATA is complete from 1963, the time of the first use of renal replacement therapy in Australia and New Zealand, containing data on all patients for whom indefinite renal replacement therapy is anticipated (and so who have no expectation of recovery of native kidney function) starting from the time of their first renal replacement treatment.

Detailed demographic, clinical and treatment data, past cancer and new incident cancer diagnoses are recorded. Cancer records are coded for site and type using adapted International Classification of Diseases for Oncology codes and where possible are supported by pathology reports. The registry complies with the Australian Commonwealth Privacy Act governing health data collection, and only data with personal identifiers removed are released for analysis, so maintaining patient anonymity. Quality control methods for data entry and consistency are routine for all data held in ANZDATA, and corrections are also made when inaccuracies are identified when the data are used. Further detailed information about ANZDATA can be found on the registry website www.anzdata.org.au

For this study we defined cancer as any neoplasm that would normally require reporting under statutory law to the cancer registries in Australia and New Zealand; we excluded nonmelanocytic skin cancer, pre-malignant and *in situ* lesions and lip cancers. Regrettably, we could not include basal and squamous cell skin cancers or pre-malignant and *in situ* lesions at other sites as these are not recorded for the general population, so without available incidence data we could make no comparison with the transplant recipient cases. We excluded lip cancers, as recent ANZDATA audit has suggested that cancers at this site may have been systematically under-recorded (19). Patients were censored at date of death or last reported contact, but were not censored at the time of transplant failure as we reasoned that cancer risk was likely to persist despite graft failure, previous work has established that dialysis is also associated with increased cancer risk over the general population, and because the aim of our study was to inform clinicians and patients about cancer risks at the time of decision making and listing for kidney transplantation when graft loss and reversion to dialysis are a clinical reality (13,20,21).

Statistical methods

Rates of cancer were compared to the normal population, and standardized rate ratios (SRR with 95% confidence intervals) calculated, first for all cancers, and then by site for the most common cancers, using indirect standardization, by age, sex, calendar year and country. Australian general population incidence data were obtained from the Australian Institute of Health and Welfare, and for New Zealand were obtained from the New Zealand Health Information Service, and were only available from 1980–2002, so analyses were restricted to prevalent transplant recipients with new incident cancers occurring within the same years 1980–2002 to avoid potential bias (22).

Within the transplanted population, risk factors and absolute risk of cancer were modeled using survival analysis methodology for all patients registered on ANZDATA from 1963–2004 with patients included from the date of transplantation until the time of their first post-transplant cancer diagnosis, death or last contact, whichever occurred first.

Cancer risk factors considered within the transplanted population were limited to unmodifiable factors known to clinicians and patients at the time of transplantation; age, sex, donor-kidney source, primary underlying cause of ESKD, year of transplantation, prior time spent on dialysis, self-reported racial background, history of prior nonskin malignancy and peak panel reactive antibody (PRA) measurement. Age, time on dialysis and donor-kidney source were categorized, and era of transplantation was examined as a continuous variable and then dichotomized at 1985, the year of introduction of the calcineurin inhibitor cyclosporin and near uniform change in immunosuppression therapy in Australia and New Zealand. Cause of ESKD was divided into three groups: glomerulonephritis/IgA nephropathy, diabetes mellitus and other causes, because previous published data have suggested that those with glomerular causes of ESKD are at greater risk, and those with diabetes at lesser risk of post-transplant cancers (14,23,24). The effect of graft failure on persisting cancer risk was modeled using a categorical time-dependent covariate from the date of return to dialysis.

Probabilities of cancer were calculated using Cox proportional hazard regression, giving hazard ratios (HR with 95% CI). Variables demonstrating an association with post-transplant cancer with $p < 0.10$ on univariate analysis were entered into a multivariate model. The least significant variable using the likelihood ratio (LR) test was removed sequentially until only variables with $p < 0.05$ remained. Potential effect modification was assessed using two-way interaction terms, with $p < 0.05$ regarded as significant. The proportional hazards assumption was checked using the time-dependent covariate method and plots of Schoenfeld residuals. The final Cox regression model was used to calculate the adjusted HR and to tabulate the expected proportion of patients with cancer for different patient sub-groups. Cumulative predicted cancer risk was illustrated graphically by plotting the adjusted failure function of the Cox regression. An 'expected' curve for the general population was derived using the life-table method, by applying the conditional probabilities of cancer for an equivalent general population undergoing ageing, and plotting a locally weighted regression smoothed (lowess) curve of the cumulative hazard. All analyses were undertaken using Stata software version 8.2 (StataCorp LP, College Station, TX).

Results

From 1963 to December 2004, a total of 15 183 people had received a kidney transplant in Australia (12 728 recipients) and New Zealand (2455 recipients), of whom 1640 (10.8%) developed at least one cancer, with 386 cancers occurring after graft failure. Of the total cohort, 8550 (56.3%) were alive, 6479 (42.7%) had died, 154 (1.0%) had been lost to follow-up, primarily through emigration outside Australia and New Zealand. The median duration of follow-up per patient was 7.2 years, inter-quartile range (IQR) 2.7–13.6 years (mean 9.0 years, SD 7.5), representing a total of 130 186 person-years of observation. The median duration of graft function following transplantation was 5.7 years, IQR 0.9–10.1 years (mean 6.7, SD 6.6). Table 1 shows the characteristics of the total recipient population.

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Table 1: Characteristics of kidney transplant recipients with unadjusted risks of cancer

Characteristic	All recipients	Recipients with cancer	Unadjusted HR (95% CI)	p*
Total	15 183	1640		
Age at transplantation				<0.0001
<35	5141	434	1.00	
35–44	3393	371	1.84 (1.60–2.12)	
45–54	3762	437	2.69 (2.35–3.08)	
≥55	2887	398	4.57 (3.96–5.27)	
Sex				0.002
Female	6249	738	1.00	
Male	8934	902	0.86 (0.78–0.94)	
Donor source				<0.0001
Cadaveric	12 138	1459	1.00	
Living unrelated	612	15	0.67 (0.57–0.78)	
Living related	2433	166		
Era of transplantation				0.001
Before 1985	5032	753	1.00	
1985 and after	10 151	887	1.20 (1.07–1.33)	
Primary renal disease				0.009
Glomerulonephritis/IgA nephropathy	6745	735	1.00	
Other causes	7284	854	1.08 (0.97–1.19)	
Diabetes mellitus	1154	51	0.72 (0.54–0.96)	
Time on dialysis prior to 1st transplant				0.008
<6 months	3907	440	1.00	
6 months < 1 year	2955	357	1.04 (0.90–1.19)	
1 < 2 years	3585	412	1.17 (1.02–1.34)	
≥2 years	4736	431	1.23 (1.08–1.41)	
Racial background				<0.0001
Non-white	1969	109	1.00	
White	13 214	1531	1.50 (1.24–1.83)	
Prior malignancy				<0.0001
None	14 882	1596	1.00	
Pre-transplant cancer	301	44	2.08 (1.54–2.80)	
Peak PRA at transplantation†				0.44
<50%	12 108	1399	1.00	
≥50%	1471	168	1.07 (0.91–1.25)	
Graft status during follow-up				<0.0001
Graft functioning	10 211	1254	1.00	
Graft failed	4972	386	0.74 (0.65–0.83)	

*p value quoted is for likelihood ratio X^2.

†PRA = panel reactive antibody, a measure of auto-immune sensitisation. Peak PRA measurements were not available for all patients as PRA have only been recorded by ANZDATA since 1990. Analysis of this risk factor is limited to 13 579 patients.

To estimate the excess risk of cancer compared to the general population, 12 633 transplant recipients prevalent between 1980–2002 were analyzed, representing 104 084 years at risk, during which time 1324 (10.5%) developed at least one new cancer. For both sexes and at all ages, the rates of cancer were elevated compared to those observed in the general population, and these are illustrated in Figures 1A and B. A female transplant recipient aged 25 experienced a rate of cancer equivalent to a 55-year-old woman in the general population (779.2/100 000 compared with 776.8/100 000, respectively), and a 45 year old female recipient experienced a rate equivalent to a 70-year-old female (1458.9/100 000 compared with 1408.7/100 000, respectively). For males the situation was similar, with 30-year-old recipients experiencing a rate of 435.5/100 000, similar to a 50-year-old male in the general population (458.8/100 000). Overall, women with kidney transplants experienced three times more cancers than women in the general population (SRR 3.19; 95% CI 2.94–3.45), while for men risk was two and a half fold (SRR 2.58 95% CI 2.40–2.78). This excess relative risk was not constant but was inversely related to age: the greatest relative risk was experienced by younger recipients of both sexes, with risk declining towards that experienced by the general population with increasing age (Table 2 and Figure 2).

Risk also varied by cancer site. Table 3 shows relative risk of cancers by age and sex for the most common cancers in the general population (colorectal, lung, melanoma,

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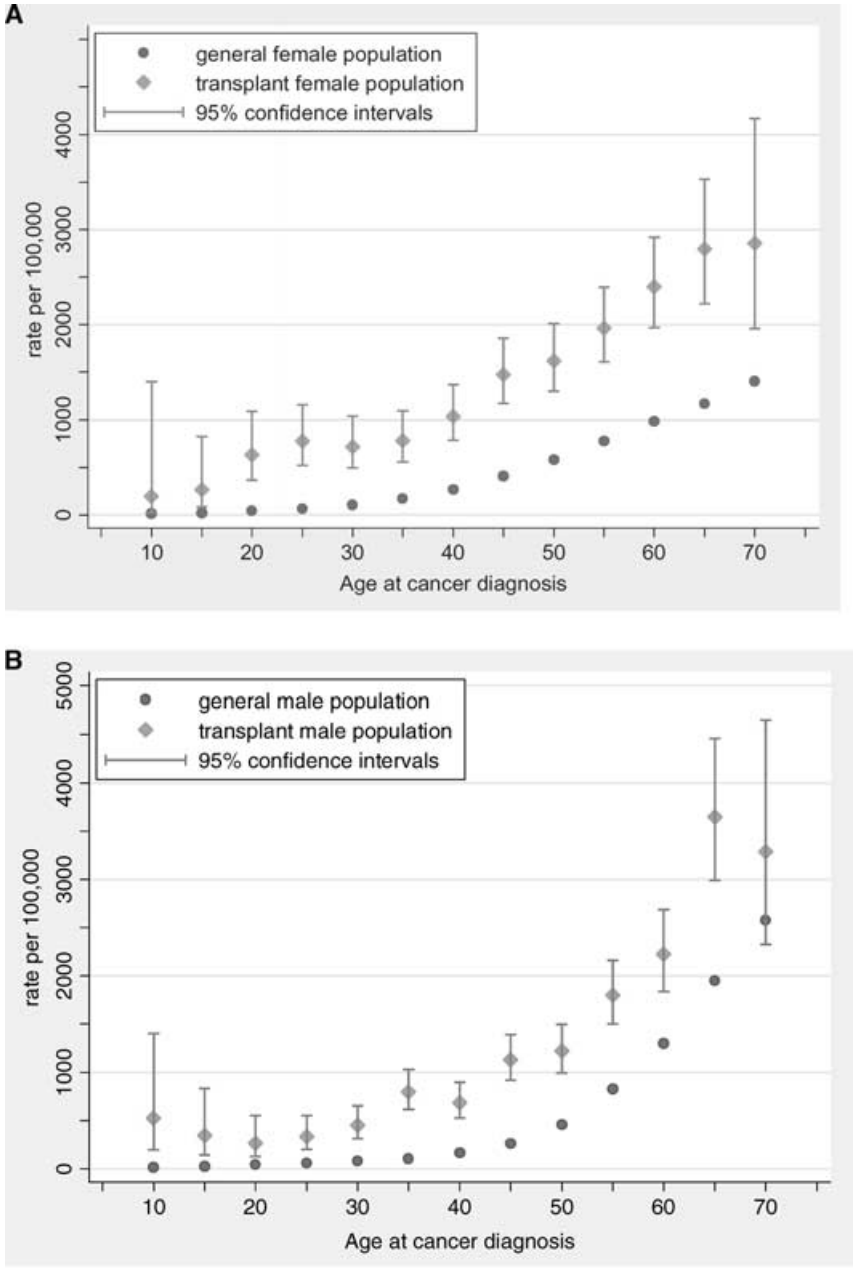


Figure 1: Incident cancer rates for women (A) and men (B) in Australia and New Zealand with a kidney transplant, compared to the age-matched general population.*

*General population data obtained from Australian Institute of Health and Welfare and the New Zealand Health Information Service, by age, sex and calendar year of diagnosis.

breast and prostate) and the most common cancer in the transplant population: lymphoma (classified in the general and transplanted populations to include Hodgkin's and non-Hodgkin's lymphoma) (25). For colorectal cancer in both sexes, risk followed the observed overall age trend and was the greatest for younger patients, falling with age (SRR 13.51 and 2.26 in females <35 years and ≥55 years respectively) with similar trend for breast and lung cancer. The same pattern occurred for lymphoma, although the magnitude of risk remained considerably higher (SRR 23.03 and 6.61 for male recipients <35 and ≥55 years, respectively). In contrast, elevated risk of melanoma for males and females showed less variability across ages, and prostate cancer was not significantly increased at all.

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Table 2: Overall relative risk of cancer diagnosis for ANZDATA posttransplant population compared to the general population of Australia and New Zealand

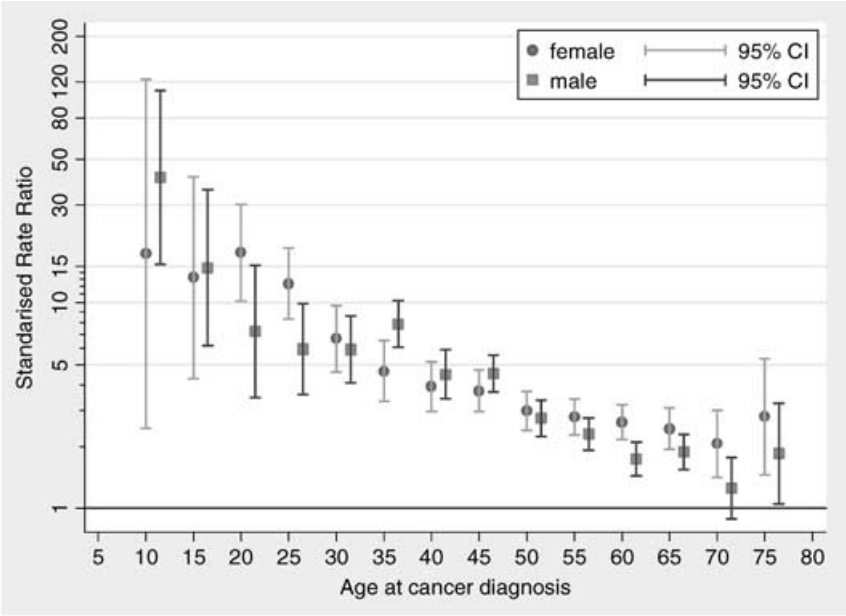
Age at cancer diagnosis		Recipients	Years at risk	Observed cancer	Expected cancer	SRR (95% CI)*
All ages	Female	5192	42712.41	610	191.44	3.19 (2.94–3.45)
	Male	7441	61371.95	714	281.47	2.58 (2.40–2.78)
<35	Female	1712	10995.65	71	7.26	9.88 (7.83–12.47)
	Male	2519	16021.51	60	8.83	6.86 (5.33–8.84)
35–44	Female	1926	9081.861	83	19.81	4.20 (3.39–5.21)
	Male	3035	14898.94	110	19.55	5.78 (4.79–6.96)
45–54	Female	2247	9822.45	152	46.11	3.31 (2.82–3.88)
	Male	3392	15421.64	181	54.52	3.41 (2.95–3.95)
≥55	Female	2111	12812.46	304	118.26	2.56 (2.29–2.87)
	Male	2857	15029.87	363	198.57	1.86 (1.67–2.06)

*Standardized rate ratio calculated by indirect standardization by country, age, sex and calendar year, 1980–2002.

Within the transplant population, to identify which unmodifiable factors at the time of transplant conferred greater risk of subsequent malignancy, initial unadjusted analyses suggested that all variables were associated with significantly increased risk of cancer except peak PRA prior to transplantation ($p = 0.44$) (Table 1). After allowing for the effects of the other variables, time on dialysis prior to transplantation ($p = 0.49$), era ($p = 0.38$) and donor source ($p = 0.25$) were no longer significantly associated with post-transplant cancer, and not analyzed further. There was an evidence that, while age at transplantation and sex both significantly affected cancer risk, there was also an interaction so that risk was different for males and females, but the magnitude of the difference changed as age at transplantation increased (interaction $p = 0.007$).

The final predictive model contained age at transplantation, sex, an interaction term between age at transplantation and sex ($p = 0.007$), primary kidney disease ($p = 0.01$), racial background ($p = 0.001$), prior malignancy ($p = 0.04$) and the time dependent covariate for graft failure ($p = 0.03$). The proportional hazards assumption held for all variables within the final model and for the model itself (global test $p = 0.15$).

The adjusted risks of cancer after kidney transplantation are displayed in Table 4, with risk estimates by age at transplantation shown separately for males and females. Women were significantly more likely than men to be diagnosed with a cancer if transplanted before the age of 45 (males vs. females <45 years HR 0.76; 95% CI 0.62–0.93).



*General population data obtained from Australian Institute of Health and Welfare and the New Zealand Health Information Service, by age, sex and calendar year of diagnosis.

Figure 2: Standardised rate ratios for incident cancers by age and sex, for kidney transplant recipients in Australia and New Zealand (1980–2002).*

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Table 3: Site-specific cancer risk for kidney transplant recipients, by age and sex

Age at cancer diagnosis	<35 years			35–44 years			45–54 years			≥55 years		
	O*	E*	SRR (95% CI)	O*	E*	SRR (95% CI)	O*	E*	SRR (95% CI)	O*	E*	SRR (95% CI)
Female												
Breast	4	1.28	3.12 (1.17, 8.31)	11	7.80	1.41 (0.78, 2.51)	30	19.23	1.56 (1.08, 2.21)	34	37.31	0.91 (0.65, 1.27)
Colorectal	3	0.22	13.51 (4.34, 41.61)	8	1.16	6.88 (3.44, 13.75)	18	4.91	3.66 (2.31, 5.82)	45	19.88	2.26 (1.69, 3.03)
Melanoma	6	1.90	3.17 (1.42, 7.04)	7	3.25	2.46 (1.23, 4.93)	18	4.90	3.88 (2.47, 6.08)	33	8.93	3.70 (2.63, 5.20)
Lung	0	0.04	.	0	0.38	.	8	1.97	4.06 (2.03, 8.11)	30	10.08	2.98 (2.08, 4.26)
Lymphoma†	19	0.51	37.30 (23.79, 58.48)	11	0.75	14.67 (8.13, 26.50)	16	1.61	9.95 (6.09, 16.24)	33	5.23	6.30 (4.48, 8.87)
Male												
Prostate	0	0.00	.	0	0.13	.	2	4.02	0.50 (0.12, 1.98)	43	47.87	0.90 (0.64, 1.20)
Colorectal	0	0.33	.	13	1.93	6.73 (3.91, 11.60)	11	8.67	1.27 (0.70, 2.29)	37	33.14	1.12 (0.81, 1.54)
Lung	1	0.08	11.81 (1.66, 83.83)	4	0.80	5.03 (1.89, 13.39)	13	5.45	2.2.39 (1.39, 4.11)	48	27.85	1.72 (1.30, 2.29)
Melanoma	10	2.13	4.69 (2.53, 8.72)	20	4.57	4.38 (2.82, 6.79)	23	8.34	2.74 (1.82, 4.12)	46	15.54	3.15 (2.38, 4.17)
Lymphoma†	25	1.09	23.03 (15.56, 34.09)	23	1.85	12.43 (8.26, 18.71)	31	3.42	9.06 (6.37, 12.88)	50	7.57	6.61 (5.01, 8.72)

*O = observed incident cancers, in ANZDATA cohort, between 1980 and 2002; E = expected number of incident cancers in Australian and New Zealand general population of the same age and sex distribution, occurring over the same calendar years.
†Lymphoma classified in the general and transplanted populations to include Hodgkin's and non-Hodgkin's lymphoma. Comparison is made only with incidence in the Australian general population, as comparable Lymphoma data was not available for New Zealand.

Between the age of 45 and 54 there was no sex difference, and after the age of 55 men were more likely to be diagnosed with a cancer although their increased risk was not statistically significant (males vs. females ≥55 years HR 1.13; 95% CI 0.93–1.39). For both men and women, increasing age at transplantation was associated with escalating risk of post-transplant cancer, but the risk for men increased more dramatically as age at transplantation increased.

Table 4: Risk factors for cancer within the kidney transplant population

Characteristic[1]	Hazard ratio (95% CI)
Age at transplantation (males)	
<35	1.00
35–44	1.81 (1.49, 2.19)
45–54	2.98 (2.48, 3.57)
≥55	5.18 (4.29, 6.25)
Age at transplantation (females)	
<35	1.00
35–44	1.82 (1.49, 2.24)
45–54	2.24 (1.83, 2.74)
≥55	3.50 (2.82, 4.34)
Male versus female at age	
<35	0.77 (0.63, 0.93)
35–44	0.76 (0.62, 0.93)
45–54	1.02 (0.84, 1.23)
≥55	1.13 (0.93, 1.39)
Primary renal disease	
Glomerulonephritis/IgA nephropathy	1.00
Other	0.93 (0.84, 1.03)
Diabetes mellitus	0.67 (0.50, 0.89)
Racial background	
Non-white	1.00
White	1.36 (1.12, 1.65)
Prior malignancy	
No cancer history	1.00
Prior cancer	1.40 (1.03, 1.89)
Graft status at cancer diagnosis	
Graft functioning	1.00
After graft failure	0.88 (0.78, 0.99)

[1]Results shown are from the final multivariate cox model adjusted for age at transplantation, sex, an interaction term between age at transplantation and sex, primary kidney disease, racial background, prior malignancy and the time-dependent covariate for graft function.

There was no significant difference between cancer risk for recipients whose primary kidney disease was caused by glomerulonephritis or other causes, but risk was significantly reduced for those who had diabetic ESKD (HR 0.67; 95% CI 0.50–0.89). White racial background conferred a similar increased risk for developing cancer as the history of a prior malignancy (HR 1.36 and HR 1.40, respectively). Graft failure and return to dialysis significantly reduced risk of subsequent cancer (HR 0.88) (Table 4).

The predicted cumulative risk of cancer for males and females from time of transplantation are represented graphically by age group in Figure 3, and are adjusted for the effects of primary disease, race, prior cancer and the possibility of graft failure. An expected cumulative risk in an Australian and New Zealand general population of the same age is also shown. Women <45 years in the general population experience more cancer than men, but in the transplant population the sex difference is more pronounced. For all age groups, the predicted curves remain appreciably above the expected and continue to diverge with time from an otherwise comparable ageing general population. Recipients aged 55 and over, being a selected group of patients who have undergone extensive physical examination prior to transplantation, initially experience less cancer than would be expected in the general population of the same age for the initial 3 years after transplantation, before the increased malignancy risk conferred by transplantation drives their predicted incidence above that background incidence. Although in this group transplanted men experience more cancer than transplanted women, the difference in not significant, whereas in the general population this sex difference is marked among older people.

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Discussion

Summary of key findings

This study described the epidemiology of cancer (excluding nonmelanocytic skin and lip cancer) in 15 183 kidney recipients transplanted over a 40-year period, followed for a median of 7.2 years. Compared with the cancer experience of an equivalent general population, the relative risk experienced by children with transplants was many times higher than those without transplants, and although the magnitude of risk declined with increasing age (Figure 2), for recipients over 65 the risk was still 2–3 times above that of the general population.

Rates of cancer in transplant recipients are roughly equivalent to rates experienced by people 20–30 years older without transplants (Figures 1A and B). In absolute terms, the excess burden of cancer cases in the transplant population compared with the general population varied from 184.6/100 000 extra cases in girls aged 10–14 (where 197.8 were observed compared to the 13.2/100 000 in the general population) to 1656.2/100 000 extra cases in men of 65–69 (where 3604.8 were observed compared to 1948.6/100 000 expected) (Figures 1A and B).

The site-specific cancer risk for transplant recipients also differed from that seen in the general population: although

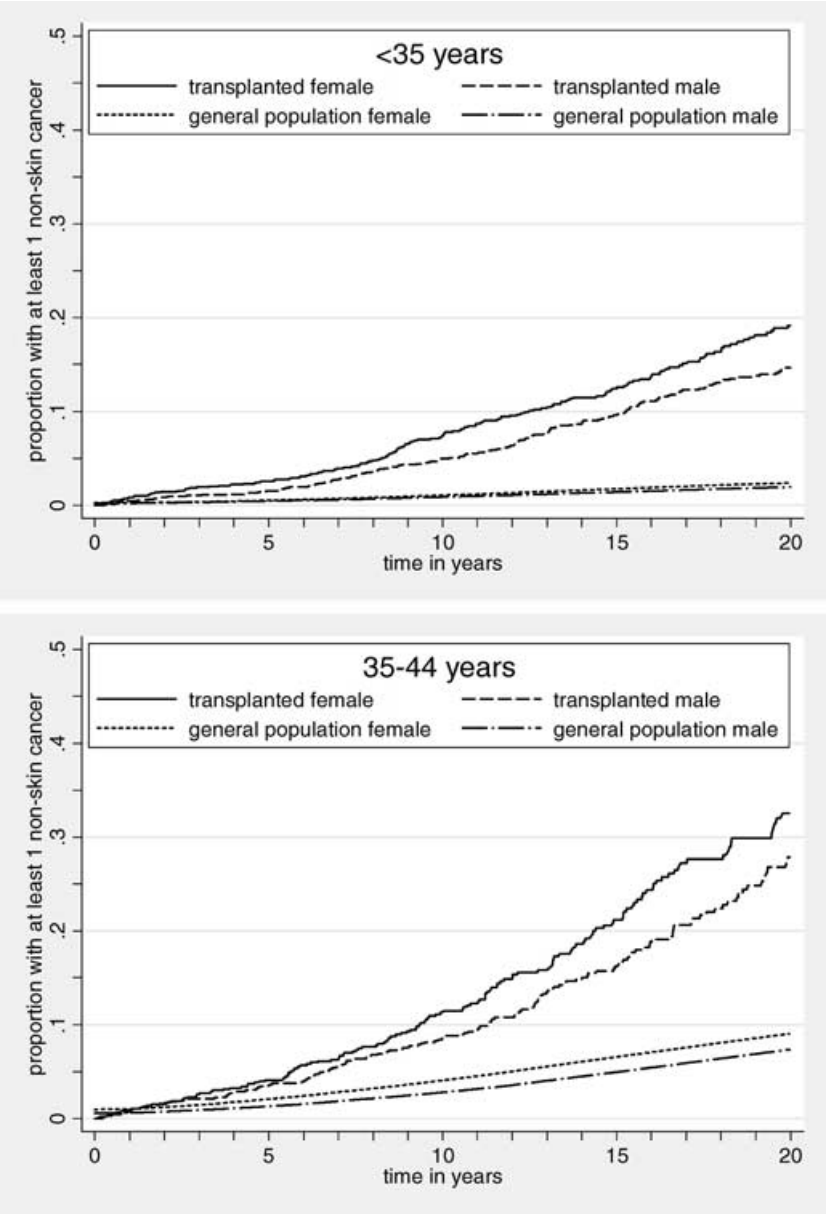


Figure 3: Cumulative risk of cancer (excluding nonmelanocytic skin and lip cancer) in kidney transplant recipients by age at transplantation, with expected cumulative risk for a comparable general population of the same age and sex.

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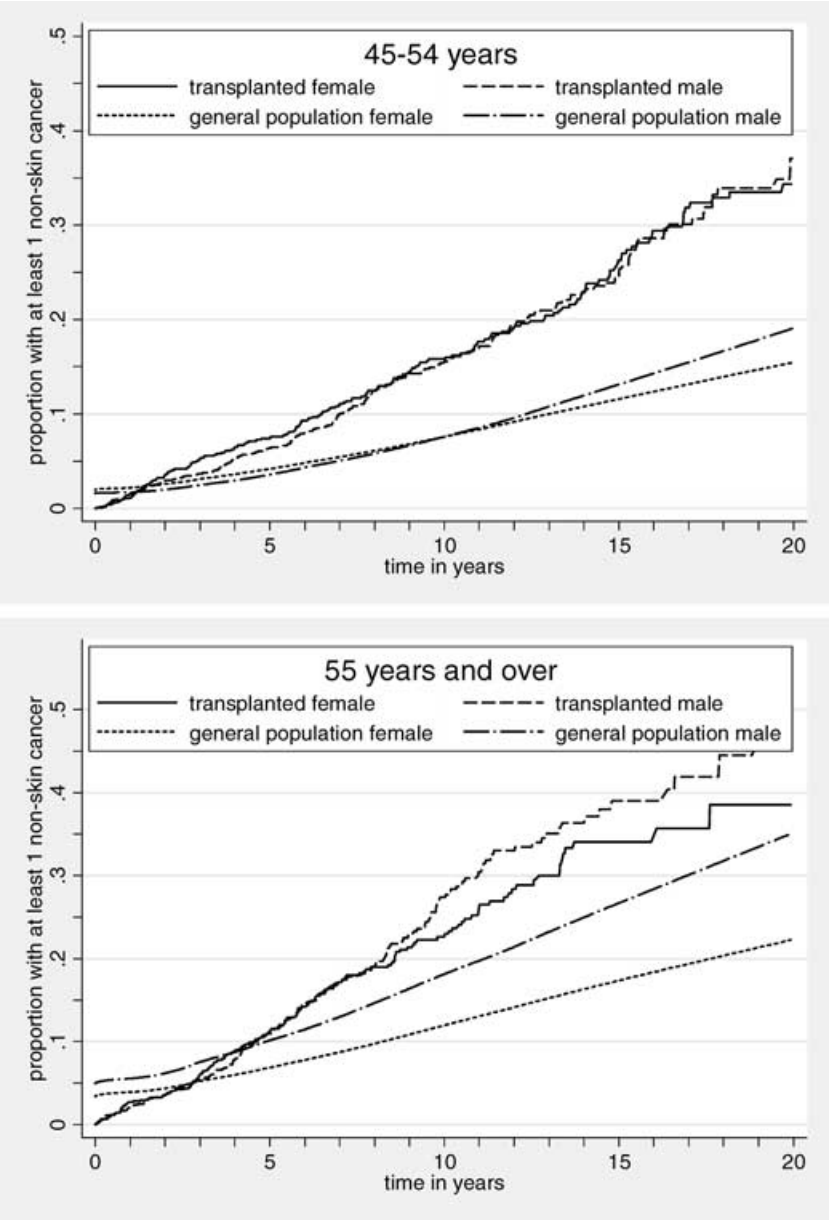


Figure 3: Continued.

more cases than expected were observed of the commonest cancers for all ages (except for breast and prostate cancers in those ≥55 years), the relative risks (SRR) varied in magnitude both by site and across age groups (Table 3).

Risk factors for cancer within the transplanted population showed that young women have a 23% greater risk than young men transplanted at the same age, but this sex difference is eliminated for those aged 45–54, where the risk is comparable, and then reversed for the older age group, where men have increased risk (Table 4). When expected cumulative risk is illustrated (Figure 3), the sex difference is partially, but not wholly explained by the expected sex differences in the general population.

Summary of findings in the context of published literature

The major strength of this analysis is that it was based on a large cohort of patients with a longer duration of follow-up than any previous published studies (13,14,16). Additionally, the prospective nature of ANZDATA data collection, the population-based coverage and the completeness of the data fields mean that the registry provides a unique dataset with selection and ascertainment biases minimized. Although globally there are transplant registries with larger numbers of patients, they are less complete in the geographical area covered, and rely on linkage with external databases for cancer and other data (14,16,17,26,27).

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Table 5: Reference table showing absolute risk of a cancer diagnosis: expected cases per 100 kidney recipients at 1, 5 and 10 years after transplantation for different patient groups

Primary renal disease	Racial background	Prior cancer history	Graft function	Age at transplantation < 35 years						Age at transplantation 35–44 years					
				1		5		10		1		5		10	
				F	M	F	M	F	M	F	M	F	M	F	M
GN/IgA	White	No	Yes	0.7	0.5	3.0	2.1	7.3	5.2	1.2	0.8	5.4	3.6	12.7	9.5
			Failed	0.6	0.4	2.7	1.9	6.4	4.8	1.1	0.8	4.7	3.4	11.2	8.5
		Cancer	Yes	0.9	0.7	4.2	3.0	10.0	7.5	1.2	1.2	7.4	5.3	17.3	13.2
			Failed	0.8	0.6	3.7	2.6	8.8	6.6	1.5	1.1	6.5	4.7	15.3	11.7
	Non-white	No	Yes	0.5	0.4	2.1	1.6	5.2	4.0	0.8	0.6	3.7	2.8	9.3	7.2
			Failed	0.4	0.3	2.0	1.3	4.7	3.4	0.8	0.5	3.5	2.5	8.4	6.4
		Cancer	Yes	0.7	0.5	3.1	2.2	7.5	5.6	1.3	0.9	5.5	4.0	13.0	9.9
			Failed	0.6	0.4	2.7	1.9	6.6	4.9	1.1	0.5	4.8	3.5	11.5	8.8
Other	White	No	Yes	0.6	0.4	2.8	2.0	6.8	4.9	1.1	0.8	5.0	3.6	11.9	9.0
			Failed	0.6	0.4	2.5	1.7	6.0	4.4	1.0	0.7	4.4	3.1	10.4	7.9
		Cancer	Yes	0.9	0.6	3.9	2.8	9.4	7.0	1.6	1.1	6.9	4.9	16.2	12.3
			Failed	0.7	0.5	3.4	2.4	8.2	6.1	1.4	0.9	6.1	4.4	14.3	10.9
	Non-white	No	Yes	0.5	0.3	2.1	1.5	5.0	3.7	0.8	0.6	3.7	2.6	8.9	6.7
			Failed	0.4	0.3	1.8	1.3	4.4	3.3	0.7	0.5	3.2	2.3	7.8	5.9
		Cancer	Yes	0.7	0.5	2.9	2.0	7.0	5.2	1.2	0.8	5.1	3.7	12.2	9.2
			Failed	0.6	0.4	2.5	1.8	6.1	4.6	1.0	0.7	4.5	3.2	10.7	8.1
DM	White	No	Yes	0.5	0.3	2.1	1.4	5.0	3.6	0.8	0.6	3.7	2.6	8.8	6.6
			Failed	0.4	0.3	1.8	1.3	4.4	3.2	0.7	0.5	3.2	2.3	7.7	5.8
		Cancer	Yes	0.6	0.5	2.9	2.0	6.9	5.1	1.1	0.8	5.1	3.6	12.0	9.1
			Failed	0.6	0.4	2.5	1.8	6.0	4.5	1.0	0.7	4.4	3.2	10.6	8.0
	Non-white	No	Yes	0.3	0.2	1.5	1.1	3.7	2.7	0.6	0.4	2.7	1.9	6.5	4.9
			Failed	0.3	0.2	1.3	0.9	3.2	2.4	0.5	0.4	2.4	1.7	5.7	4.3
		Cancer	Yes	0.5	0.3	2.1	1.5	5.1	3.8	0.8	0.6	3.8	2.7	9.0	6.8
			Failed	0.4	0.3	1.8	1.3	4.5	3.3	0.7	0.5	3.3	2.4	7.9	6.0

Primary renal disease	Racial background	Prior cancer history	Graft function	Age at transplantation 45–54 years						Age at transplantation ≥ 55 years					
				1		5		10		1		5		10	
				F	M	F	M	F	M	F	M	F	M	F	M
GN/IgA	White	No	Yes	1.5	1.4	6.5	6.2	15.2	15.4	2.2	2.5	9.7	10.6	22.3	25.1
			Failed	1.3	1.3	5.7	5.5	13.4	13.6	2.0	2.2	8.6	9.3	19.8	22.4
		Cancer	Yes	2.1	2.0	9.0	8.6	20.6	20.8	3.1	3.4	20.3	14.0	29.7	33.2
			Failed	1.8	1.7	7.9	7.6	18.3	18.4	2.7	3.0	11.8	12.9	26.5	30.1
	Non-white	No	Yes	1.1	1.1	4.8	4.6	11.4	11.6	1.7	1.8	7.2	7.9	16.9	19.2
			Failed	1.0	0.9	4.2	4.1	10.1	10.2	1.4	1.6	6.4	7.0	14.9	17.1
		Cancer	Yes	1.5	1.4	6.7	6.4	15.6	15.8	2.3	2.5	10.0	10.8	22.8	25.7
			Failed	1.3	1.3	5.8	5.6	13.8	14.0	2.0	2.2	8.8	9.6	20.2	23.1
Other	White	No	Yes	1.4	1.3	6.0	5.8	14.2	14.3	2.1	2.3	9.1	9.8	20.9	23.5
			Failed	1.2	1.1	5.3	5.1	12.5	12.7	1.8	2.0	8.0	8.7	18.5	21.1
		Cancer	Yes	1.9	1.8	8.3	8.0	19.3	19.4	2.9	3.2	12.5	13.4	27.9	31.2
			Failed	1.7	1.6	7.3	7.1	17.1	17.2	2.5	2.8	11.0	11.9	24.9	28.2
	Non-white	No	Yes	1.0	1.0	4.5	4.3	10.7	10.8	1.5	1.7	6.7	7.3	15.8	17.9
			Failed	0.9	0.9	3.9	3.8	9.4	9.5	0.3	1.5	5.9	6.5	14.0	15.9
		Cancer	Yes	1.4	1.4	6.2	6.0	14.6	14.7	2.1	2.3	9.3	10.1	21.4	24.1
			Failed	1.2	1.2	5.4	5.2	12.9	13.0	1.9	2.0	8.2	8.9	19.0	21.6
DM	White	No	Yes	1.0	1.0	4.4	4.2	10.5	10.6	1.5	1.6	6.7	7.2	15.6	17.7
			Failed	0.9	0.8	3.9	3.7	9.3	9.4	0.3	1.5	5.9	6.4	13.8	15.7
		Cancer	Yes	1.4	1.3	6.1	5.9	14.4	14.5	2.1	2.3	9.2	9.9	21.1	23.9
			Failed	1.2	1.2	5.4	5.2	12.7	12.8	1.9	2.0	8.1	8.8	18.7	21.4
	Non-white	No	Yes	0.7	0.7	3.3	3.1	7.9	7.9	1.1	1.2	4.9	5.4	11.7	13.3
			Failed	0.6	0.6	2.9	2.8	6.9	7.0	1.0	1.1	4.3	4.7	10.3	11.9
		Cancer	Yes	1.0	1.0	4.5	4.4	10.8	10.9	1.6	1.7	6.8	7.4	16.0	18.1
			Failed	0.9	0.9	4.0	3.8	9.5	9.6	1.4	1.5	6.0	6.5	14.1	16.2

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Our finding of accelerated risk with increasing age at transplantation is in agreement with other published literature; however, other authors have found that male sex confers higher risk overall. This may be explained by our analytic approach: despite the recognized importance of analysis of sex differences in epidemiological studies, particularly in cancer epidemiology (28,29) to our knowledge the interaction between the effects of age at transplantation and sex on cancer risk has never previously been explicitly assessed (13,14,16,23,27,30–32).

That those whose ESKD was caused by diabetes experienced 32% less cancer than those with ESKD from other causes remains incompletely explained, and our interpretation is guarded. The reduction in cancer risk for diabetics has been noted in at least three other published analyses and never been adequately explained (14,23,30), and is in contrast to recent cohort studies in the general population, which have suggested that those with diabetes may be at greater risk of certain cancers (33). Despite our efforts, our analysis techniques may not have adequately adjusted for early censoring in this group, who may be disproportionately more likely to experience the 'competing risk' of dying of other causes and complications of ESKD and consequently be less likely to develop cancer. Additionally, our measurement of the effect of diabetes may be confounded as we have not captured those recipients with ESKD from causes other than diabetes who also have or subsequently develop diabetes as comorbidity. Potential further investigation of the 'diabetes effect' will be limited by the difficulty in sourcing comparable data of diabetes comorbidity among the general population.

The influence of race on cancer risk in transplant recipients has been noted in analyses from American registries who demonstrate decreased risk in African-Americans and Asians relative to American whites (14,16). In Australia and New Zealand, the population mix differs, with indigenous Australians, Mäori and Pacific Islanders contributing to the non-white population. These racial groups may have reduced incidence of melanoma, colon, breast and prostate cancers compared to the general population (34), and within the transplanted population also have reduced graft and overall survival, both of which may contribute to our findings (35). That race behaves as a proxy for unmeasured socio-economic or environmental factors that may confound cancer risk is also possible, but our ability to investigate these with available data was limited. Linking ANZDATA to external data sources might allow fuller exploration of this association so that the basis of the observed excess risk for whites can be better determined.

Implications for patient care

There has been much recent debate about communicating risk to clinicians and how best to convey risk to patients, which has emphasized the need for good quality information, graphics and the use of absolute and not relative risks where possible (36–38). In addition to Figures 1A and B, the estimates presented in Table 5 will provide patients with a greater understanding of the likelihood of being diagnosed with malignancy after transplantation through a clear risk estimate appropriate to their individual situation, rather than the more generalized estimates available until now. Where 1 in 14 white women transplanted before the age of 35 with glomerulonephritis causing ESKD, no prior history of a cancer and a graft functioning for 10 years can expect a cancer, this rises to 1 in 3 white men with a prior history of cancer transplanted aged 55 or over, but with ESKD from the same cause. Similarly, men aged 45–54 at transplantation with graft function at 10 years have cancer risks varying from one in 13 (non-white, diabetic ESKD, no prior cancer) to one in five (white, prior cancer, ESKD from other causes).

A greater understanding of a recipient's cancer risk from the outset of transplantation will allow clinicians to subsequently manipulate factors that are open to modification, such as the immunosuppressive drug regimen, and future work could examine this. This study may also have implications for post-transplant care and help inform the balance of surveillance in the context of scarce resources and differing clinical practice guidelines among countries. This work also suggests that guidelines may need revision. Both European and American guidelines suggest screening for prostate cancer with prostate specific antigen (PSA) for all male transplant recipients over the age of 50 (39,40), but our analysis demonstrates comparable or reduced risk of prostate cancer compared to the general population, and although the public health debate on PSA mass-screening is unresolved, current policy does not advocate screening the general population (41,42). European and American guidelines also suggest regular mammography for recipients over 50, where mass-screening of the general population of similar age has shown mortality reduction (43). However, our analysis shows that the risks for female transplant recipients relative to the general population are greater at younger ages. A more detailed investigation into breast cancer incidence, stage, treatment and survival in the transplant population may be warranted, to assess whether screening has similar benefits for all ages, and if the decreased mortality observed in the screened general population can be assumed for those with ESKD.

Conclusions

The rates of cancer in kidney recipients are similar to non-transplanted people 20–30 years older, but within the transplanted population cancer relative and absolute risk differs across patient groups depending on age, sex, primary cause of ESKD, race, prior malignancy and continued graft function. We have provided explicit information to inform clinicians of expected risk based on patient characteristics known before transplantation, to guide counselling and subsequent post-transplant care.

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Partial preliminary reports of this analysis were presented at the XX International Congress of the Transplantation Society, Vienna 2004.

Sources of Support and Full Disclosure of Potential Conflicts of Interest

ANZDATA registry: receives financial support from several sources including government and industry. Those contributing funds have no involvement in how funds are used, or rights of authorship or publication in studies using registry data. The registry and study authors retain the right to interpretation of the results and the right to publish. Funds are held and managed in a general fund by a third party. Funding sources are/have been; the Australian Government Department of Health and Ageing, the New Zealand Department of Health, and Kidney Health Australia (formerly the Australian Kidney Foundation), Amgen Australia, Janssen-Cilag Australia, Novartis, Wyeth Australia, and Fresenius.

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JCC: none declared

JMS: none declared

MPJ: none declared

JRC: none declared

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